Quiz 08

MGTE 31423 - Advanced Optimization Methods in Management Science

1, ½ hours

- ***The marks allocated to each part are given in brackets***
- ***Late submissions will incur a penalty of 5% for every 5 minutes of delay.***
- ***Please include the student numbers of all group members who contributed to the assessment in your submission***

01. A multinational company plans to invest \$50 million in renewable energy projects across two divisions: Solar and Wind. The company can allocate any fractional amount of the budget to each division, and the expected return from each division is a continuous function of the investment. Assume that the investment decision is made sequentially, where the company first allocates funds to one division and then allocates the remaining budget to the other.

    Using the concept of infinite-state dynamic programming, answer the following:

    i. Explain why this problem is classified as an infinite-state dynamic programming problem. (2 Marks)
    ii. Clearly define the stages, states, and the objective for this problem. (3 Marks)
    iii. Explain how optimality applies in this situation using dynamic programming. (4 Marks)

    Note: You may design the DP application according to your preference. Clearly state any assumptions made in your formulation.

02. A technology startup has \$5 million to invest in improving its digital infrastructure. The company must allocate this budget across three strategic investment areas:

    1. Cybersecurity (C)
    2. AI Development (A)
    3. Cloud Infrastructure (I)

    Each investment area must receive at least \$1 million. The \$5 million can be distributed among the three areas. The estimated increase in annual profit (in \$ million) depending on the amount invested is given in the table.

Investment (\$M)	Cybersecurity	AI Development	Cloud Infrastructure
1	5	4	3
2	9	7	6
3	12	9	8
4	14	10	9

5	15	10	9

Define Stages, States and Objective. Using dynamic programming to allocate \$5 million investment among Cybersecurity, AI Development, and Cloud Infrastructure to maximize the total annual profit increment. State all the possible optimal solutions. (15 Marks)

03. A rescue drone can carry at most 7 kg of emergency supplies. There are 5 supply kits available. The corresponding weights and survival values are given in the table.

The goal is to select kits that:

- Do not exceed 7 kg
- Maximize total survival value

Kit	Weight (kg)	Survival Value
1 (Water Pack)	4	20
2 (Medical Kit)	3	18
3 (Food Pack)	5	25
4 (Thermal Blanket)	2	10
5 (Satellite Phone)	3	15

Determine the maximum survival value and which kits are selected. (15 Marks)